REACH

RULES & GUIDELINES

presented by **PROTARA PRINTING**

1. What is this?

This is a global CAD design competition where engineers and designers of all backgrounds come together to compete in computer-aided design. With a $250 prize pool and a real-world engineering challenge, it's your chance to showcase your CAD skills against talented designers worldwide.

Whether you're a seasoned CAD professional or just starting out, this competition is an opportunity to test your design abilities, sharpen your technique, and connect with a community of passionate builders. All you need is CAD software and your creativity. **Entry is completely free.**

2. The Challenge

This month's game is titled REACH, and it will test your ability to design a desktop robotic arm in CAD.
Your objective is to model a 3 degree-of-freedom (3 DOF) desktop robot arm built around a provided set of Arduino-driven components. You may use any CAD software of your choice (Fusion 360, SolidWorks, Onshape, FreeCAD, Inventor, Solid Edge, or anything else that can export a neutral file format.)

A reference Onshape document will be provided showing the Arduino parts, motors, power source, and supporting hardware you should design around. You don't have to model these parts from scratch — they're there so your arm is built to actually house and drive real components.

Submissions are graded on CAD quality, aesthetics, functionality, feasibility, and uniqueness. Bonus points are awarded for a creative and unique function your arm can perform, and for clean, well-organized CAD. You may also submit an optional portfolio (Google Doc or presentation) alongside your model for additional points — see the Portfolio Bonus rubric.

3. Eligibility

•Open to everyone, of all ages, in any country. Entry is free.
•Entrants under 18 must have a parent or guardian provide consent at registration. Entrants under 13 must register through a parent or guardian.
•You may enter solo or as a small team (up to 3 people). Team entries split any prize.
•One entry per person or team.
•Protara team members, helpers, and event judges are not eligible to win prizes.

4. How to Enter?

1. Register through the official form linked on the website (**free**).
2. Read these rules in full and review the reference Onshape document.
3. Design your 3 DOF desktop robot arm in the CAD software of your choice.
4. Submit your deliverables (below) before the deadline.

5. Provided Component Reference

The Onshape reference document includes the following parts. Your arm should be designed to physically house, mount, and move using this component set. You are free to add to it, but a competitive entry should clearly accommodate these. Advanced entrants may swap servos for NEMA 17 stepper motors with A4988 / DRV8825 drivers and a CNC shield for higher precision and torque. This is welcome and counts toward feasibility, but the 3 DOF requirement and all other rules still apply.

#	Component	Role in the build
1	Arduino Uno R3	Main microcontroller running the arm
2	3× MG996R Servo Motor	The three actuated joints. One per DOF
3	PCA9685 16-channel PWM	Drives the servo motors to let the arm move
4	Hiwonder 6V 5A	Power supply to run the whole arm

6. The Three Joints (DOF)

Your model is required to have at least three degrees of freedom (DOF). It doesn't strictly need to follow the layout below, but we'll list out the typical joints as reference.

•**Joint 1** — Base rotation (waist): rotates the whole arm; high-torque servo, often paired with a base bearing for smooth motion.
•**Joint 2** — Shoulder: lifts the arm; the highest-load joint.
•**Joint 3** — Elbow: extends the reach; lighter-load servo.

Optional / Enhancement Components

Not required, but they make for a stronger, more functional entry:

- Gripper servo (SG90 / MG90S) for an end effector - note this adds a 4th DOF beyond the required 3.
- 3× potentiometers or an analog joystick for manual control input.
- Base ball bearing for smooth, low-friction waist rotation (great to model).
- Servo brackets / horns - often the structural backbone of the arm.
- Mounting clamp to fix the arm to a surface or corner
- Gearboxes to multiply servo motor torque.

7. Submission Requirements

To keep judging fair across every CAD platform, all submissions must include:

Required

•A STEP file (.stp / .step) of your full arm assembly — the universal format every judge can open. This is the file that gets judged.
•2–3 rendered images or screenshots showing your arm from multiple angles.
•A short component list (BOM) noting which provided Arduino components your design uses and where.
•Your native CAD file (.f3d, .sldprt, Onshape link, etc.). (Protara Staff will wipe names from all files before forwarding to judges)
Ensure your name or any identifying information isn't present any deliverables.

Optional (recommended)

•A turntable GIF or short clip of the arm rotating.
•An optional portfolio, a Google Doc or slide deck documenting your design (graded separately for bonus points).

Originality

Your design must be your own original work created for this competition. Submitting a pre-made arm from GrabCAD, Thingiverse, or any library is grounds for disqualification. We may ask for in-progress screenshots or version history as proof of work.

8. Judging Rubric

Every submission is scored out of 100 points on the five core criteria, with up to 30 additional bonus points available through the bonus, and the portfolio. Judges score each criterion on a 1–5 scale, which is then averaged and multiplied with the critereon weighting. For example, and average grade of 3.5 on Functionality would earn 14 points, while the same grade on Aesthetics would earn 10.5 points.

Criterion	Weight	What's being scored
CAD Quality & Execution	x5 (25 points)	Clean geometry, a properly constrained assembly, correct mates/joints, no errors or floating parts, parametric/editable intent
Feasibility & Engineering Soundness	x4 (20 points	Would it actually work? Realistic joints and load paths, sensible wall thickness, correctly accommodates the provided components
Functionality	x4 (20 points)	Achieves a true 3 DOF, sensible range of motion, the joints and any end effector do what they're meant to
Aesthetics & Presentation	x3 (15 points)	Visual design of the arm, render quality, and how clearly the entry is communicated
Uniqueness & Innovation	x4 (20 points)	Originality of the design, mechanisms, or approach. Does it stand out from a generic arm?

9. Bonus Points – 10 points

Bonus	Max	Awarded for
Creative/unique function	+8	A genuinely clever or unexpected thing the arm can do beyond basic reach-and-grab
Clean CAD Organization	+2	Named features, organized feature tree, logical folders, named mates/joints, tidy assembly structure

10. Portfolio Bonus – 20 points

If an entrant submits the optional portfolio doc/presentation, it is scored separately for extra points and never penalizes anyone who skips it. **Ensure your name or any identifying information isn't present on any deliverables.**

Bonus	Max	Awarded for
Design process & rationale	+6	Explains the why behind key design decisions, and/or shows iteration and thought process
Presentation polish	+6	Turntable GIF, exploded view, motion study, or standout renders
Completeness (BOM + specs)	+4	Clear component list, dimensions, and how it all fits together
Clarity & communication	+4	Well-organized and readable, with helpful visuals (diagrams, drawings, etc.)

11. Scoring Anchors

•**1 — Poor**: largely missing, broken, or unconsidered.
•**3 — Solid**: meets expectations; competent and complete.
•**5 — Excellent**: exceptional; clearly above standard. (Judges may use 2 and 4 for in-between work.)

12. Judging Process & Integrity

1. The panel consists of 2-3 neutral judges who neither Protara Staff nor entrants.
2. Entries are judged anonymously. Entrant names are removed by Protara staff before scoring by judges.
3. Judges with any personal connection to an entrant recuse themselves from scoring that entry (conflict-of-interest rule).
4. Tie-breaks are resolved by the highest Feasibility & Engineering Soundness score, then highest CAD Quality.
5. Judges' decisions are final, but feedback forms and rubrics are available at request.

13. Prize Distribution

Once judging is concluded, winners will be announced by email, and on Protara's instagram page and website. For this month's cadathon, the total prize pool is **$250**

- **1**st place — $125 Protara Printing Credit
- **2**nd place — $80 Protara Printing Credit
- **3**rd place — $45 Protara Printing Credit

Prize split within the $250 is subject to final confirmation and may be adjusted see protaraprinting.com/cadathons for the confirmed breakdown.

14. Event Timeline

Milestone	Date
CADathon topic revealed	July 1st, 2026
Submissions open	July 1st, 2026
Submission deadline	July 15th, 2026
Winners announced	July 17th, 2026

15. Code of Conduct

This is a friendly, global, all-ages community. By entering, you agree to:

- Submit only your own original work.
- Keep all communication respectful and inclusive.
- Not harass, plagiarize, or attempt to game the judging process in any way, shape, or form.

Violations may result in disqualification at the discretion of Protara staff. Protara Printing LLC reserves the right to invalidate any fraudulent, inappropriate, or troll submissions.

16. Legal & Terms

Intellectual property: You retain ownership of your design. By entering, you grant Protara Printing LLC a non-exclusive, royalty-free license to display, reproduce, and share your submission (with credit) for promoting Cadathon REACH and Protara across our website, social media, and event materials. You confirm your entry is your own original work and does not infringe anyone else's rights.
Consent to use likeness: Winners and participants agree that their name, username, country, and submitted work may be published in connection with the competition. Entrants under 18 require guardian consent for this, collected at registration.
Privacy & data: We collect only the information needed to run the event (name, email, country, age, and guardian details for minors). We do not sell your data or share it beyond what is required to administer the competition and award prizes. Minor data is handled with extra care in line with applicable child-privacy laws.
Prizes: Prizes are awarded as described in the form of gift card codes to Protara Printing's website.
Liability: Cadathon REACH is a design competition; no physical hardware is built or shipped as part of entry. Protara Printing LLC is not liable for any loss, damage, or technical issue (lost files, software errors, connectivity) arising from participation, to the fullest extent permitted by law.
Conduct & disqualification: We reserve the right to disqualify any entry for plagiarism, rule violations, or misconduct, at our sole discretion.
Changes: Protara Printing LLC reserves the right to modify, postpone, or cancel the competition, or amend these rules, with notice posted at protaraprinting.com. Protara staff and judges' decisions are final.
Agreement: By registering, you confirm you have read and agree to these rules and terms in full.

Questions?

Reach out via our email, or Instagram We'll keep a running FAQ on the competition page at *protaraprinting.com/cadathon*.

Email: *support@protaraprinting.com*
Instagram: protara.printing

BEST OF LUCK, AND HAPPY CADDING

– THE PROTARA TEAM